

No. of Functional Features

HDI Prediction System — Project Development Phase

Overview

The HDI Prediction System includes a focused set of functional features that together deliver the complete prediction workflow, from user input to result display.

Functional Feature List

#	Functional Feature	Description
1	Indicator Input Form	Collects Life Expectancy, Schooling years, and GNI per Capita
2	Input Validation	Ensures all four fields are filled before submission
3	HDI Prediction Engine	Runs the trained Linear Regression model on submitted inputs
4	Result Display	Shows the predicted HDI score on a dark-themed results page
5	Dark-Themed UI	Consistent navy/dark styling across form and result pages
6	Responsive Layout	Interface adapts to different screen sizes
7	Model Persistence	Trained model loaded once and reused for all predictions
8	Public Deployment	Application hosted and accessible via Render.com

Feature Count Summary

Total functional features implemented: 8. These cover the complete end-to-end flow — data input, validation, prediction, result display, styling, responsiveness, model efficiency, and public accessibility.